

VISHAL MALLARAPU

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Full Stack Engineer specialized in Serverless Architecture, AWS ROSA, and Insurance Domain

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 6+ years of experience designing, building, and operating large-scale enterprise AWS Cloud platforms. Deep expertise in dual-stack architectures which are built for Insurance domain for Home and Auto Insurance over 5 years using Guidewire, Gosu lang and many more ranging from containerized Java Spring Boot microservices on AWS ROSA to event-driven Node.js serverless workflows. Proven track record of migrating legacy TP2 monoliths to cloud-native environments and implementing robust SRE practices including chaos engineering and observability. Experienced in Agile delivery, mentoring junior engineers, and driving DevOps maturity in highly regulated financial environments.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Spring MVC, Node.js, JavaScript, TypeScript, Angular, REST APIs
AWS Serverless: Lambda, API Gateway, DynamoDB, EventBridge, S3, Secrets Manager, SSM Parameter Store, STS, IAM
GenAI Tools: Gemini Pro, ChatGPT, Claude Sonnet, Git Co-Pilot, Figma AI
Container Platform: AWS ROSA (OpenShift), Kubernetes, Docker, Helm, OpenShift Routes, HPA
DevOps & IaC: Terraform, GitLab CI, Argo Rollouts, Renovate, Semantic Release, Maven, Artifactory, GitOps
Testing & Quality: Karate (API), Gatling (Perf), JUnit, Jest, Sinon, Chaos Engineering, Snyk (Security), SonarQube
Observability: Splunk, Dynatrace, Prometheus, CloudWatch, Logback
Security & Auth: OAuth 2.0, MSAL, HashiCorp Vault, JWT, Merna, Corporate Proxy
Databases & Messaging: PostgreSQL, IBM MQ, Hibernate/JPA, NoSQL (DynamoDB)

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (Client: State Farm)

Software Engineer - Full Stack

Charlotte, NC

2021 – Present

Project 1: Utility Services (Enterprise Microservices Platform)

- **Legacy Migration (TP2 to AWS ROSA):** Led the strategic migration of 3 core applications and 4 backend services from legacy TP2 systems to AWS ROSA. Containerized Java Spring Boot microservices using Docker and successfully integrated them with PostgreSQL and IBM MQ for asynchronous processing.
- **Guidewire Insurance Domain:** Have experience in dealing with Guidewire architecture in turn getting Data from Policies and their migrations over 15 State Group Rollouts for a project which spans over 8 years.
- **Lender Relations (TPFI) & Token Management:** Developed and maintained critical utility services supporting Lender Relations (TPFI), Driver License Validation, and Sensitive Token Data Management. Ensured 99.99% availability for these shared services which underpin the entire Auto Policy Center.
- **Location & Object Services:** Built specialized location services responsible for Address Validation, Location Rating, and Earthquake Zone detection. Engineered the "Physical Object ID" service to create and link identifiers to vehicles, ensuring data integrity across the policy lifecycle.
- **Progressive Delivery & DevOps:** Implemented GitOps methodologies using GitLab CI and **Argo Rollouts** for blue-green and canary deployments. Automated dependency updates using **Renovate** and versioning via **Semantic Release**, reducing deployment lead time by 40%.
- **Performance & Reliability Testing:** Validated system throughput under peak loads using **Gatling** for performance testing and **Karate** for API automation. Integrated **Chaos Engineering** experiments to test resiliency against pod failures and latency injection.
- **Security & Compliance:** Enforced security standards using **Snyk** for vulnerability scanning and **Merna** for secrets management. Implemented JWT-based authentication to secure inter-service communication.

Project 2: Event Management System (Serverless Architecture)

- **Event-Driven Architecture:** Architected a serverless eSignature orchestration system using **Node.js**, **AWS Lambda**, and **Amazon EventBridge**. Designed the system to listen for specific triggers, automatically generating signature packages and routing customers to DocuSign.

- **Webhooks & Idempotency:** Engineered a robust webhook handler to process callback events from DocuSign. Implemented idempotency checks using **DynamoDB** to prevent duplicate processing of financial documents and resolved contingencies automatically upon signature completion.
- **Infrastructure as Code (IaC):** Provisioned all AWS resources including API Gateway, Lambda, and DynamoDB using **Terraform**. Managed configuration secrets securely using **AWS Secrets Manager** and **SSM Parameter Store**.
- **Frontend Integration:** Developed Angular UI components to guide users through the eSignature flow, implementing **Jest** and **Sinon** for unit and mock testing of frontend logic.
- **Authentication Strategy:** Implemented secure authentication flows using **MSAL**, **OAuth 2.0**, and **HashiCorp Vault**, ensuring compliant access control for sensitive customer data.

DietOnClick.com

Junior Software Engineer

Hyderabad, India

2018 – 2019

- **Full Stack Development:** Developed and enhanced core web application features using Java, Spring MVC, and JSP. Collaborated closely with UI/UX teams to translate wireframes into responsive, functional interfaces.
- **Database Optimization:** Designed and optimized MySQL database schemas to support user profile management and diet tracking features. Wrote complex SQL queries and improved data retrieval speeds for high-traffic pages.
- **Frontend Modernization:** Revamped legacy JavaScript codebases, introducing modular scripts to improve maintainability and browser compatibility across devices.
- **Quality Assurance:** Partnered with QA teams during release cycles to identify and patch bugs, ensuring a stable deployment process for the customer-facing portal.

HCLTech

Software Engineering Intern

Chennai, India

April 2014 – Sep 2014

- **Cricket Divisional Database Project:** Collaborated within an Agile development team to build a local cricket divisional scores database and live update portal for a regional client.
- **Web Development Training:** Gained hands-on experience in full SDLC fundamentals, web development standards, and relational database management systems.
- **Team Collaboration:** Participated in daily stand-ups and sprint planning sessions, learning the fundamentals of Agile methodology and iterative delivery in a professional client-facing environment.
- **Feature Delivery:** Assisted in the development of the live score update module, implementing data refresh logic to provide real-time match statistics to end-users.

University of North Carolina at Charlotte

Graduate Teaching Assistant (Computer Science)

Charlotte, NC

2020 – 2021

- Mentored undergraduate students in Java, Data Structures, and Algorithms. Conducted code reviews, debugging sessions, and graded assignments, reinforcing core software engineering principles.

EDUCATION

University of North Carolina at Charlotte

Master of Science in Information Technology (GPA: 3.65/4.0)

Charlotte, NC

May 2021

- *Relevant coursework: Human Computer Interaction and Interfaces, Network-Based App Development, Database Systems, Agent based Intelligent Systems, Algorithms & Data Structures*

Vellore Institute of Technology

Bachelor of Technology in Computer Science (GPA: 7.5/10)

Vellore, India

Nov 2018

CERTIFICATIONS & LEADERSHIP

Certifications: AWS Certified AI Cloud Practitioner, Google Developer Launchpad (NodeJS, Python, Java and Git Cloud)

Leadership: General Secretary, College of Computing and Informatics - UNC Charlotte